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Comparison of Symmetrical Prism Adaptation to Asymmetrical Prism Adaptation in those with Normal Binocular Vision

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Abstract

This study sought to determine whether symmetrical compared to asymmetrical horizontal prisms (base-out or base-in) evoked different rates of phoria adaptation. Sixteen young adults with normal binocular vision participated in a symmetrical phoria adaptation experiment using a 3Δ base-out or 3Δ base-in binocular prism flipper and an asymmetrical phoria adaptation experiment using a 6Δ base-out or 6Δ base-in monocular wedge prism. The experiments were randomized and counterbalanced to reduce the influence of the prism stimulation order. Asymmetrical base-out prism adaptation was significantly faster than symmetrical prism adaptation for subjects with normal binocular vision. Asymmetrical phoria adaptation with base-in prism was not significantly different from symmetrical phoria adaptation implying that there are directional asymmetries (convergent versus divergent eye movements) in the slow fusional component of vergence. Data suggest that a potential interaction between the version system and the slow fusional vergence system may exist. Results have clinical relevance because patients with convergence or divergence insufficiency / excess may potentially show more pronounced differences between symmetrical and asymmetrical phoria adaptation compared to binocularly normal controls. These differences might also be relevant to clinical measurements such as vergence fusional range, which can be measured symmetrically (with Risley prisms in a phoropter) or asymmetrically (with prism bar).

Keywords

Phoria adaptation; prism adaptation; vergence adaptation; heterophoria

1.0 Introduction

Schor describes the disparity vergence system to be composed of “fast” and “slow” fusional vergence components (Schor, 1979). While these two components are mainly driven by retinal disparity (Horwood & Riddell, 2008; McLin, Schor, & Kruger, 1988; Schor, 1979; Semmlow & Wetzell, 1979), each component is described to have different characteristics;

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specifically, each has different neural time constants. For example, if an individual with normal binocular vision is presented with a visual target, the person can use the relatively quick and accurate “fast” disconjugate movements of the eyes to reduce the disparity between the current vergence angle of the eyes and that of the target of interest. The “slow” fusional vergence component has a much longer time constant compared to the “fast” fusional vergence component and is used to slowly adapt to near or far visual space. While other depth cues are present, slow fusional vergence, can be assessed by the dissociated phoria.

The dissociated phoria is the relative ocular rotation of the eyes during binocular fixation on an object in the absence of a fusible stimulus (such as when one eye is occluded). The rotation of the eye can be eso (inward), exo (outward), ortho (no movement), hyper (upward), or hypo (downward). If a person performs sustained fixation on a target (either near or far) for 30 seconds or more, the phoria level controlled by the slow fusional vergence component will be shifted towards where the visual system’s gaze is located (Ying & Zee, 2006). For example, people who perform near work such as reading for a prolonged period of time will experience an esophoric shift in their phoria (Sreenivasan, Irving, & Bobier, 2012). This shift in phoria level reduces the effort it takes to maintain a given vergence angle (Schor, 1983). The interaction between the fast and slow fusional components maintain single binocular vision of targets that are at different locations in depth.

In addition to sustained fixation, the slow and fast fusional components can be altered by using a prism or a lens (Scheiman & Wick, 2014). If a prism is placed in front of one or both eyes, it shifts light and hence the image to a different point along the retina. When prisms are placed base-out, the eyes will rotate inward (convergent movement). Conversely, prisms that are placed base-in will evoke divergent eye rotation. Convergent and divergent responses of the slow fusional system have been found to have asymmetries and thus should be studied individually (Erkelens & Bobier, 2017; Erkelens, Thompson, & Bobier, 2016). When single binocular vision is allowed for one second (Larson & Faubert, 1994) to 15 seconds (Schor, 1979), the slow fusional vergence component begins to adapt to the power of the prism, and over time, the phoria returns to the level reported before the prism was first placed in front of the eye (Fogt & Toole, 2001; Henson & North, 1980; Sethi & North, 1987). This adaptive response is called phoria or prism adaptation. Plus or minus lenses have also been used to stimulate phoria adaptation (Sreenivasan, Irving, & Bobier, 2009).

Most phoria adaptation studies have been conducted asymmetrically: that is a single prism is placed over the right or the left eye so the field of view is shifted eccentrically along the retina of a single eye. This configuration is believed to stimulate both the version and vergence systems (Enright, 1992). However, phoria adaptation can also occur symmetrically along the midline (the subject’s midsagittal plane) when disparity is introduced by placing two prisms with the same prismatic power in front of each eye (symmetrical phoria adaptation). It is not fully understood whether these two types of phoria adaptations (symmetrical compared to asymmetrical) produce different rates of adaptation in individuals with normal binocular vision.

Controversy exists in the literature. In the laboratory, it is possible to stimulate the saccadic or vergence systems independently by carefully presenting visual stimuli that move laterally or in depth, respectively. However, under natural conditions, both systems are usually stimulated simultaneously. There are two competing theories to describe how conditions that stimulate the integration of vergence and saccadic eye movements occur. One theory is the additivity hypothesis which supports that saccades and vergence interact nonlinearly (Coubard, 2013). On the other hand, the Ditchburn hypothesis supports independent saccadic control of each eye that is minimally influenced by the vergence system (Enright, 1996). Numerous research papers show experimental evidence that support either theory depending on the experimental conditions (Coubard, 2013). For asymmetrical target configurations, different amounts of rotation are required from each eye. Prior research supports that asymmetrical vergence movements are usually accompanied with saccadic eye movements (Enright, 1992). Symmetrical target stimuli presented along the subject's midline theoretically should stimulate symmetrical vergence eye movements. However, saccades and asymmetries between the disconjugate rotation assessed as differences between the left and right peak velocities have been reported in eye movements from symmetrical stimuli presented along midline for binocularly normal control subjects (Alkan, Biswal, Taylor, & Alvarez, 2011; Kim & Alvarez, 2012; Semmlow, Alvarez, & Pedrono, 2007; Semmlow, Chen, Granger, Donnetti, & Alvarez, 2009). Asymmetrical differences between the left and right eye are also more pronounced in patients with convergence insufficiency (Alvarez & Kim, 2013).

Taking into account the differences in eye movements elicited by asymmetrical and symmetrical target configurations, one plausible outcome of this study is that the rate of adaptation induced by the asymmetrical condition is faster than that of the symmetrical condition. This may potentially occur because the asymmetrical condition is a combination of vergence and version (saccadic) eye movements, which may utilize some nonlinear interaction as described by the additivity hypothesis. There is evidence showing that there are interactions and dependencies between version and vergence eye movements (Alvarez, Jaswal, Gohel, & Biswal, 2014; Erkelens, Steinman, & Collewijn, 1989; Kim, Vicci, Granger-Donetti, & Alvarez, 2011; Zee, Fitzgibbon, & Optican, 1992). Specifically, these studies suggest that saccades facilitate the peak velocity of disparity vergence. There is also recent research showing that the phoria level of an individual influences vergence peak velocity (Alvarez, 2015; Alvarez et al., 2010; Kim et al., 2011; Lee, Granger-Donetti, Chang, & Alvarez, 2009; Talasan, Scheiman, Li, & Alvarez, 2016). It may be that the asymmetrical phoria adaptation stimulation recruits the version system in addition to the vergence systems. This may lead to an increase in the rate of phoria adaptation potentially because version (saccades) facilitate vergence peak velocity. However, there is also contradicting evidence suggesting that vergence and version are independent (Alvarez et al., 2009; Alvarez, Semmlow, Ciuffreda, Gayed, & Granger-Donetti, 2007; Kim, Vicci, Han, & Alvarez, 2011; King & Zhou, 1995; Rashbass & Westheimer, 1961; Semmlow, Chen, Granger, Donnetti, & Alvarez, 2009; Semmlow, Yuan, & Alvarez, 1998). Therefore, a second plausible outcome is that the symmetrical condition could produce a faster adaptation rate since it mostly evokes pure vergence movements. Activating one system might be less complicated and time consuming than pooling resources from two systems.

Thus, the symmetrical condition might produce a faster phoria adaptation. Third, there may be no difference in the results between symmetrical or asymmetrical phoria adaptation. One of the main goals of this experiment is to determine whether symmetrical and asymmetrical phoria adaptation have similar or different adaptation rates, and if one is different which one is faster. Such knowledge has potential clinical implications, which will be described within the discussion.

2.0 Methods

Subjects

All sixteen subjects were young adults (6 males and 10 females) and were not aware of the purpose of the experiment. Their age ranged between 18 and 22 years (*Mean (M)* = 19.1, *Standard Deviation (SD)* = 1.5). The study was approved by the New Jersey Institute of Technology Institutional Review Board, and it is in accordance with the Declaration of Helsinki where subjects signed written informed consent.

Typical Vision Parameters to Assess Normal Binocular Vision

Typical vision parameters to assess binocular vision are summarized in Table 1 as a mean with one standard deviation. These measurements were recorded before subjects started the experiment to determine whether they had normal binocular vision, and could participate in the study. Only subjects with normal binocular vision who did not report being diagnosed by a clinician to have a disorder or disease that may affect vergence, accommodation or ocular motility participated in this study. All subjects had normal or corrected to normal (20/20) visual acuity. The near (40 cm) dissociated phoria was tested using the flashed Maddox rod procedure. Normal stereo vision was assessed with a Randot Stereo Test (Bernell Corp., South Bend, IN), which indicated that subjects had normal local and global binocular vision. The near point of convergence (NPC) was measured using an Accommodation Convergence Ruler (Bernell Corp., South Bend, IN) placed at the bridge of the nose using the same protocol described by the Convergence Insufficiency Treatment Trial (Convergence Insufficiency Treatment Trial Study Group, 2008). NPC break was measured in cm along midline when a target was perceived diplopic or deviation of ocular alignment to the midline target was observed. NPC recovery was also measured in cm along the midline when the subject was able to regain fusion after the NPC break. Fusional vergence range was measured using a base-in and base-out prism bar which contained 1 Δ , 2 Δ to 20 Δ in increments of 2 Δ , and 20 Δ to 45 Δ in increments of 5 Δ . The values for the vergence range for blur, break and recovery are reported in Table 1. Table 1 also reports the range of the subject's measurements, and the values that clinicians recommend to be assessed as having normal binocular vision (Scheiman & Wick, 2014).

Experimental conditions: Phoria Prism Adaptation measured with Flashed Maddox Rod Procedure

This experiment used the flashed Maddox method to measure horizontal near dissociated phoria. Subjects sat 40 cm away from a near dissociated phoria Muscle Imbalance Measure (MIM) Card placed symmetrically along the subject's midline (Bernell Corp., South Bend, IN). The phoria card was positioned at eye level for each subject by adjusting the card

vertically dependent on each subject's height. Baseline phoria was measured twice before the beginning of each of the experimental conditions to ensure that the phoria level of the subject was repeatable. The vast majority of the movements were identical and hence these measurements were averaged. For the few phoria measurements that were different, they were within 2Δ of each other and were also averaged. A difference of 2Δ or less is not considered clinically significant (Scheiman & Wick, 2014). To measure phoria, subjects looked at a penlight through the near dissociated phoria MIM card while a Maddox rod was flashed 2 to 4 times until the subject could reliably report the location of a vertical red streak of light. The subjects were encouraged to use their peripheral vision to perceive the location of the red line. The order in which the four experimental conditions (asymmetrical base-in, asymmetrical base-out, symmetrical base-in, or symmetrical base-out) were administered was counterbalanced to reduce procedural systematic bias. In the asymmetrical condition, subjects held a 6Δ wedge prism (base-in or base-out) over their right eye (see Fig.1a). In the symmetrical condition, subjects held a flipper prism with 3Δ prism (base-in or base-out) in front of each eye (see Fig. 1b). The experiment began with the subject placing the prism(s) in front of their eye(s). At this time, the researcher would immediately occlude the eye right for 15 seconds, followed by flashes of the Maddox rod. While keeping the prism(s) in front of his/her eye(s), the subject would look at a small letter chart that was located above the Phoria MIM Card for 30 seconds to stimulate prism adaptation. After the adaptation period, the near dissociated phoria was measured twice and averaged. The rest of the experimental session consisted of repeating the following steps: an adaptation period (30 seconds of binocular vision through the prism(s)), followed by occlusion of the right eye with the Maddox rod, and two phoria measurements. The two phoria measurements were done for repeatability and were averaged.

3.0 Results

The phoria measurements recorded using the Maddox rod procedure were plotted over time and fitted with mathematical exponential functions for each of the participants (Figure 2). Even though there are some individual differences, the exponential fits in Figure 2 showed that all the participants did experience adaptation to the vergence disparity produced by the prisms. The base-out prism evoked an esophoric shift in the baseline phoria level. Whereas, the base-in prism evoked an exophoric shift in the baseline phoria level. Figure 2 revealed that there were less individual differences in the asymmetrical condition with the base-out prism (Figure 2 plot c) than in the other three conditions (Figure 2 plots a, b, and d). The data from the subjects in this condition follow an exponential fit and qualitatively appeared faster than the other conditions.

The group level analysis averaged all data and then a group level exponential fit was calculated. Similar to the individual data (see Fig. 2), the mean group data (Fig. 3) showed that there were differences in the rate of phoria adaptation, especially when the asymmetrical adaptation with the base-out prism condition was compared to the other conditions. The rise of this exponential fit was faster than the other three conditions, which were more gradual. The asymmetrical base-out phoria adaptation data had a faster time constant. To quantify these findings, the adaptation rate was calculated as the differences between the final and initial phoria measurements (magnitude of the phoria adaptation) divided by the time

constant. The time constant, (τ), was the time it took for the phoria adaptation to reach 63% of steady state, or final value (see Eq. 1). The time constant is a conventional engineering parameter that is the time for the system's step response to reach 1-1/e which is approximately 63% of the steady state response. This formula has been used in prior studies to assess the velocity or rate of phoria adaptation (Alvarez, Kim, & Granger-Donetti, 2017; Alvarez, Kim, Yaramothu, & Granger-Donetti, 2017; Kim, Vicci, Granger-Donetti, & Alvarez, 2011). The velocity of the phoria adaptation rate for the asymmetrical base-out prism condition was greater (2.6 Δ /min) than the velocity of adaptation for the asymmetrical base-in prism (2 Δ /min), the symmetrical base-out (1.7 Δ /min), and base-in (1.8 Δ /min) conditions.

$$\text{Rate of adaptation} = \frac{\text{Final phoria measurement} - \text{Initial phoria measurement}}{\tau} \quad (1)$$

To test whether the differences in adaptation rates of the four conditions were significant, a repeated measure ANOVA was computed using type of phoria adaptation (symmetrical or asymmetrical), and base of prism (base-in or out) as factors. There was a significant difference between asymmetrical and symmetrical phoria adaptation ($F[1,15] = 5.3, p = 0.037$), but there was no significant difference between the phoria adaptation for base-in compared to base-out prisms ($F[1,15] = 0.97, p = 0.35$). The interaction between type of adaptation (asymmetrical or symmetrical) and base of the prism (base-in or base-out) did not have a significant effect on the rate of adaptation ($F[1, 15] = 1.46, p = 0.25$). A Fisher's Least Significant Difference (LSD) post hoc test showed a significant difference between the asymmetrical phoria adaptation with base-out prism and both of the symmetrical phoria adaptations ($p < 0.05$), but not with the asymmetrical phoria adaptation with base-in prisms ($p > 0.05$). These results indicate that the significant difference between symmetrical and asymmetrical phoria adaptation conditions is mostly driven by the relatively faster base-out asymmetrical phoria adaptation condition.

The final change in phoria level of the four experimental conditions were compared to address the question of whether one of the experimental conditions tended to produce a greater change in the final magnitude of phoria adaptation compared to the others. In other words, were participants able to completely adapt to the prism in one or more of the experimental conditions and not in others? Figure 4 shows the final change in phoria level of all of the participants of the study. Most of the participants were able to completely adapt to the 6 Δ prism after 7 minutes of binocular viewing as illustrated by the bar graphs. The mean final change in phoria for the four conditions were similar. The mean final change in phoria level for the asymmetrical phoria adaptation with base-out prism was the most complete ($M = 5.7, SD = 3.8 \Delta$) adaptation, followed by the mean final phoria level of the asymmetrical phoria adaptation with base-in prism ($M = 5.6, SD = 0.43 \Delta$). The mean final change in phoria level for the symmetrical phoria adaptation with base-in ($M = 5.5, SD = 0.71 \Delta$) and base-out ($M = 5.5, SD = 0.52 \Delta$) were very similar. The differences in mean final change in phoria level of the four conditions were not significant ($F[1, 15] = 2.2, p = 0.16$).

4.0 Discussion

Individuals with normal binocular vision exhibited adaptation in both the asymmetrical and symmetrical experimental conditions, and most subjects were able to completely adapt to the symmetrical as well as the asymmetrical prism after 7 minutes of adaptation. The group level analysis showed that asymmetrical phoria adaptation was faster than symmetrical phoria adaptation. Further statistical analysis indicated that the asymmetrical phoria adaptation with base-out prism was faster than the symmetrical phoria adaptation. The asymmetrical adaptation with base-out prism was faster than asymmetrical adaptation with base-in. It is plausible that this difference is due to convergence typically being faster than divergence (Erkelens & Bobier, 2017; Hung, Zhu, & Ciuffreda, 1997) since viewers adapting to a base-out prism are converging their eyes while those adapting to base-in prism are diverging their eyes. Previous studies have supported that there is an interaction between the vergence and the version systems, and that saccades facilitate fast fusional vergence peak velocity (Kim & Alvarez, 2012; Zee et al., 1992). For the asymmetrical experimental cases, the subjects are not using solely the vergence system to fixate on the target but are using both version and vergence. Perhaps, it is the utilization of both the vergence and version systems that leads to the increased velocity of phoria adaptation within the asymmetrical phoria adaptation cases. Eye tracking was not performed during these studies so future research is needed to determine how the eye movements differ between these experimental conditions.

In addition to basic science, this study's results have clinical implications. The phoria level is utilized to assess binocular function and for the diagnoses of many binocular dysfunctions. For example, part of the inclusion criteria and diagnosis of convergence insufficiency (CI) for the Convergence Insufficiency Treatment Trial (CITT) is having a near (40cm) dissociated phoria 4Δ more exophoric at near compared to the far (6 m) dissociated phoria. A larger phoria at near compared to far is hypothesized to lead to visual complaints because a person needs to compensate during near work (i.e. reading) for the eye's natural tendency to rotate outward (Scheiman, Gwiazda, & Li, 2011). Thus, the fast fusional vergence component of a CI patient will be stimulated more when trying to achieve single binocular vision of targets compared to a binocularly normal control. Asymmetrical phoria adaptation using a monocular prism has been studied in individuals with normal binocular vision and patients with CI. Patients with CI have slower rates of phoria adaptation than normal controls at near (40 cm) and far (6 m) distances, and with both base-in and base-out prisms (Brautaset & Jennings, 2005; North & Henson, 1992). Careful inspection of these data from CI patients show that the rate or velocity of phoria adaptation is slowest with the base-out prism at near compared to the base-in prism at near or far or the base-out prism at far. These results suggest that the slow fusional vergence components of patients with CI differs from individuals with normal binocular vision. Thus, the study of phoria adaptation for both the binocularly normal and binocular dysfunctional populations is important.

Another possible clinical implication of the present study is to potentially explain some of the differences observed between using a prism bar and Risley prisms in a phoropter to measure vergence fusional range (Thiagarajan, Lakshminarayanan, & Bobier, 2010). A prism bar is a series of discrete prisms which stimulate step changes in vergence.

Conversely, Risley prisms presented within a phoropter are continuous or gradual changes in prism which stimulate ramp changes in vergence. Prior studies speculate that the differences between these conditions were the step (discrete) compared to ramp (continuous) change in prism strength. However, the prism bar is presented to one eye (asymmetrical stimulation) whereas the Risley prisms are presented binocularly (symmetrical stimulation). Therefore, future studies may consider that if there are differences between the two methods, then those differences may also be due in part to the different adaptation rates inherent in symmetrical compared to asymmetrical presentation of a prism within each procedure.

Understanding phoria adaptation can shed light on how the slow and fast fusional vergence components inter-operate, and how deficiencies in these components may potentially lead to binocular dysfunctions such as convergence or divergence insufficiency / excess. Future studies may want to explore symmetrical and asymmetrical phoria adaptation in patients with CI since it has been found that orthoptic treatment (vision training/therapy) can increase the rate of phoria adaptation in patients with CI (Brautaset & Jennings, 2006). However, it is not fully understood how the improvement of phoria adaptation occurs and what cortical and subcortical regions are involved.

The differences between symmetrical and asymmetrical phoria adaptation found in the present experiment may be more pronounced in patients with CI compared to individuals with normal binocular vision. This may occur because CI patients have slower and more asymmetry in their fast fusional disparity eye movement dynamics compared to binocularly normal controls (Alvarez & Kim, 2013). Symmetrical phoria adaptation mostly involves pure vergence eye movements, which are known to be difficult for patients with CI (Alvarez et al., 2010; Scheiman, Talasan, Mitchell, & Alvarez, 2017). It might be useful to assess symmetrical phoria adaptation in patients with CI because asymmetrical phoria adaptation also involves the version system. Symmetrical prism adaptation maybe a better assessment to compare how functional the slow fusional vergence component is in CI patients compared to those with normal binocular vision.

Previous clinical research studying vision therapy supports that fusional range is correlated to prism adaptation (Thiagarajan et al., 2010). One potential implication of this study is that when testing the fusional vergence range of patients using a symmetrical stimulus, (i.e. using Risley prisms in a phoropter), there may be a reduction in the measured fusional vergence range values compared to fusional vergence range values measured using asymmetrical methods (i.e. prism bar). This difference may be due to a slower rate of prism adaptation that occurs with symmetrical compared to asymmetrical methods. More detailed research is needed to test this hypothesis.

5.0 Conclusions

The results of this study show that asymmetrical phoria adaptation using a base-out prism differs from symmetrical phoria adaptation suggesting an interaction between the version and vergence systems.

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Highlights

- * In phoria adaptation the version and slow fusional vergence systems may interact
- * Results have impact for measuring vergence fusional range

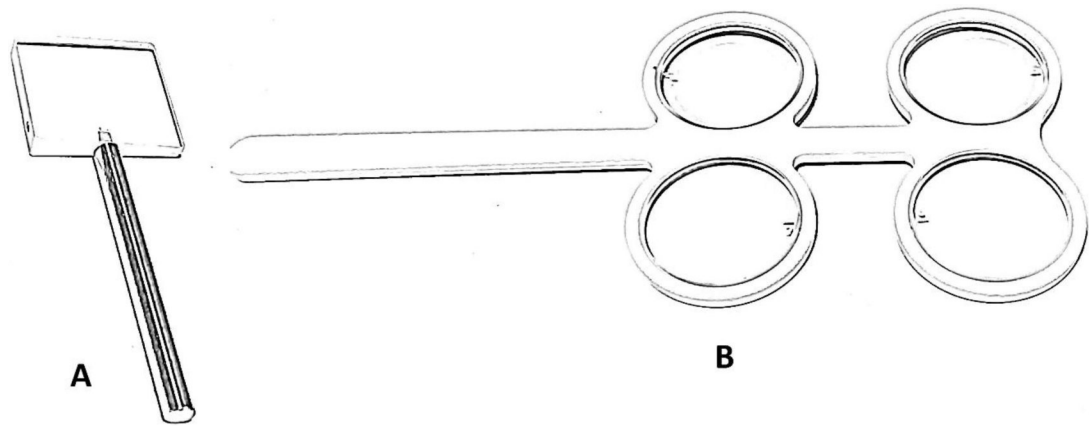


Figure 1.

The two types of prisms used to test phoria adaptation. Left-hand side (A): wedge prism used to test asymmetrical phoria adaptation. In this condition, the vergence disparity was introduced by placing a 6Δ wedge prism in front of the right eye. Right-hand side (B): flipper prisms used to test symmetrical phoria adaptation. In this condition, a vergence disparity of 3Δ was introduced to the left and to the right eye for a total vergence disparity of 6Δ .

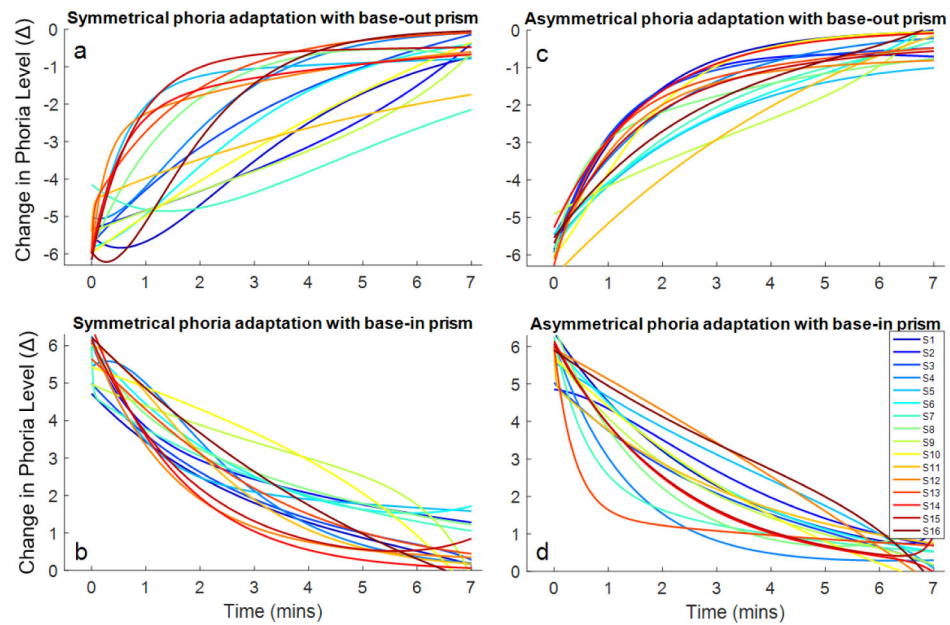


Figure 2.

Phoria measurements plotted over time (minutes) were fitted with an exponential function for each subject. The time (horizontal axis) represents the cumulative time that subjects spent using binocular vision to look through the single wedge prism or the binocular flipper prisms. The vertical axes represent the change in phoria level in Δ . Each color line represents a different subject. The four experimental conditions were symmetrical phoria adaptation (one 3 Δ prism in front of each eye) with base-out (a) and base-in prisms (b), and asymmetrical phoria adaptation (one 6 Δ prism in front of the right eye) with base-out (c) and base-in (d) vergence disparities.

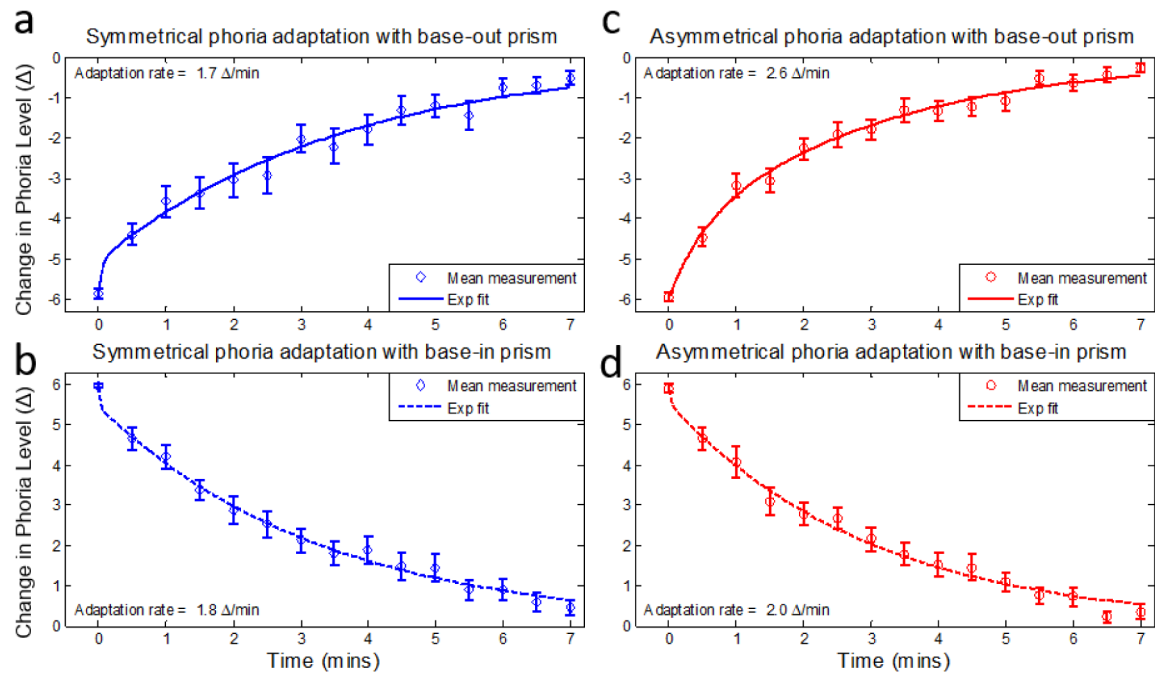


Figure 3.

Mean symmetrical (a and b) and asymmetrical (c and d) phoria adaptation in response to either base-out (a and c) or base-in (b and d) prisms. The abscissa indicates the cumulative time (in minutes) that subjects spent using binocular vision to look through the single wedge prism or binocular flipper prisms. The ordinate axes are the change in phoria level in Δ . Negative phoria levels in the ordinate axes indicates exophoria range (a and c), while positive phoria levels indicate esophoria range (b and d). Mean phoria adaptation measures were fitted with an exponential function. Error bars represent standard error (SE) of the mean.

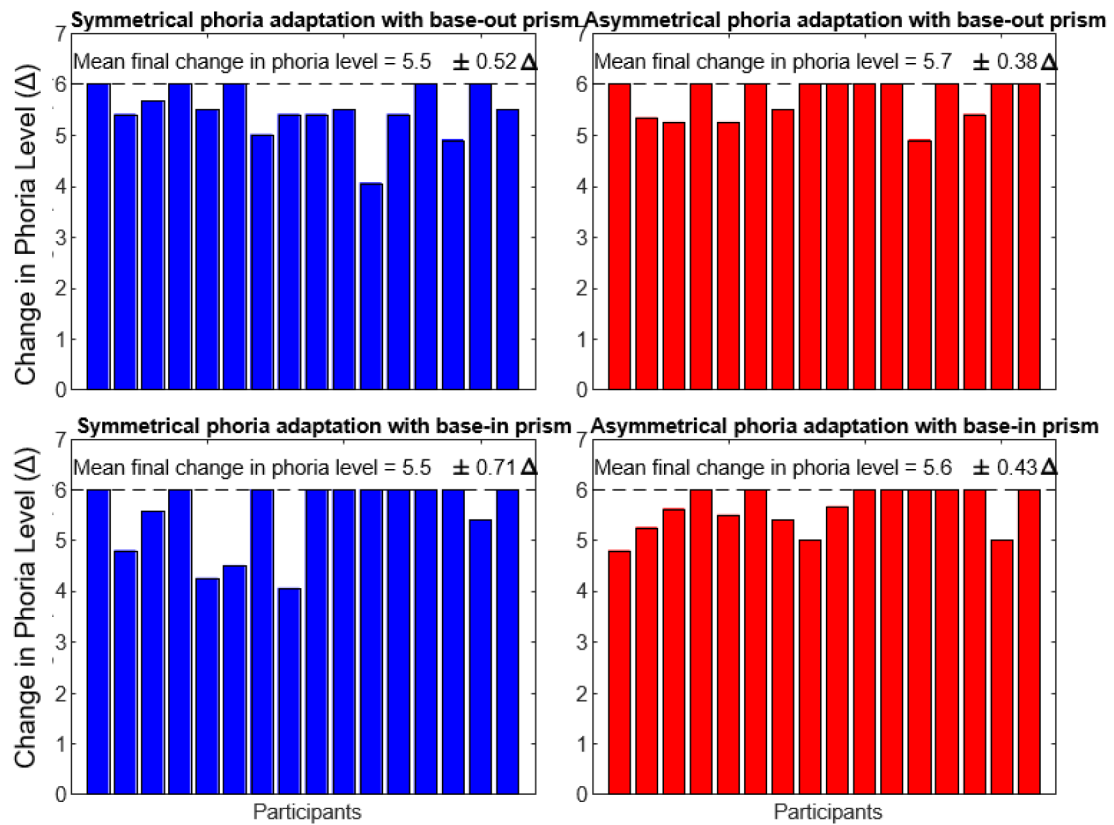


Figure 4.

Each bar in the bar graphs represent the final phoria level (Δ) of each of the participants after 7 minutes of prism adaptation. The symmetrical phoria adaptation experiment is plotted in blue and the asymmetrical phoria adaptation experiment is plotted in red. The group mean of the change in final phoria level is shown at the top of the bar graphs for the four experimental conditions with $\pm$ one standard deviation (SD). The participants who reached an adaptation of 6Δ were able to completely adapt to the prism.

Table 1.

Mean clinical measures used to test the binocular vision of all subjects prior to the start of the experiment. Right column shows expected values derived from normative data, which are used by clinicians as a guideline to assess normal binocular vision.

Baseline clinical measurements	Subjects average $\pm$ one standard deviation	Ranges (minimum to maximum values)	Reference values for normal binocular vision $\pm$ one standard deviation
Dissociated phoria at 40 cm	3.1 Δ exophoria $\pm$ 2.4	2 eso to 6 exo	3 Δ exophoria $\pm$ 3
Near point of convergence (blur)	9.1 cm $\pm$ 1.6	7 to 12 cm	
Near point of convergence (break)	4.1 cm $\pm$ 1.2	1 to 5 cm	2.5 cm $\pm$ 2.5
Near point of convergence (recovery)	6.4 cm $\pm$ 1.6	1.5 to 8 cm	4.5 cm $\pm$ 3
Local stereopsis at 40 cm	27 sec of arc $\pm$ 7	20 to 40 sec of arc	<50 sec of arc *
Global randot stereopsis at 40 cm	250 sec of arc	250 sec of arc	250 sec of arc **
Vergence range with base-out prism (blur)	17.2 Δ $\pm$ 7.1	8 to 30 Δ	
Vergence range with base-out prism (break)	25.8 Δ $\pm$ 9.9	16 to 45 Δ	19 Δ $\pm$ 9
Vergence range with base-out prism (recovery)	18.9 Δ $\pm$ 7.7	10 to 35 Δ	14 Δ $\pm$ 7
Vergence range with base-in prism (blur)	10.7 Δ $\pm$ 3.1	8 to 14 Δ	
Vergence range with base-in prism (break)	15.4 Δ $\pm$ 5.4	10 to 30 Δ	13 Δ $\pm$ 6
Vergence range with base-in prism (recovery)	12.6 Δ $\pm$ 4.4	6 to 20 Δ	10 Δ $\pm$ 5

Note: Mean vision measurements with one standard deviation. Five subjects did not report NPC to blur. Seven subjects did not report blur when their vergence range was tested with a base-out prism. Only three of the subjects reported blur when their vergence range was tested with a base-in prism.

* Normal is defined as less than 70 sec of arc.

** 250 sec of arc is the best assessment of this device